9.4 The Comparison Test

Theorem (The Direct Comparison Test): Let $\sum a_n$, $\sum c_n$, $\sum d_n$ be series with nonnegative terms. Suppose that for some integer N, we have $d_n \le a_n \le c_n$ for all $n > N$.

(a). If $\sum c_n$ converges, then $\sum a_n$ also converges.

(b). If $\sum d_n$ diverges, then $\sum a_n$ also diverges.

Proof: Consider the partial sums $S_n = a_1 + a_2 + \dots + a_n$.

Let $M = a_1 + \dots + a_N$

(a) Suppose $\sum c_n$ converges to L.

For $n > N$, $S_n = (a_1 + a_2 + \dots + a_N) + a_{N+1} + \dots + a_n$

$\le M + \sum_{k=N+1}^{n} c_k \le M + L$

S_n bounded above $\Rightarrow$ $\sum a_n$ converges.

(b) For $n > N$, $S_n \ge M + \sum_{k=N+1}^{n} d_k \to \infty$ $\Rightarrow$ $\sum d_n$ diverges $\Rightarrow S_n$ unbounded $\Rightarrow \sum a_n$ diverges.

Examples:

(a). The series $\sum_{n=1}^{\infty} \frac{5}{5n-1}$ diverges.

Note that $\frac{5}{5n-1} \geq \frac{1}{n}$ for all n

$\Rightarrow \sum \frac{1}{n}$ diverges. $\overset{\text{DCT}}{\Rightarrow} \sum \frac{5}{5n-1}$ diverges.

(b). The series $\sum_{n=1}^{\infty} \frac{1}{n!}$ converges.

Note that $\frac{1}{n!} \leq \frac{1}{n^2}$ for $n \geq 4$.

The 2-series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, so by the DCT, $\sum_{n=1}^{\infty} \frac{1}{n!}$ also converges.

(c). The following series converges.

$$5 + \frac{2}{3} + \frac{1}{7} + \underbrace{1 + \frac{1}{2+\sqrt{1}} + \frac{1}{4+\sqrt{2}} + \frac{1}{8+\sqrt{3}} + \ldots + \frac{1}{2^n+\sqrt{n}} \ldots}_{\sum_{n=0}^{\infty} \frac{1}{2^n+\sqrt{n}}}$$

We want to apply DCT, we observe that

$$\frac{1}{2^n+\sqrt{n}} \leq \frac{1}{2^n} \quad \text{for all } n \geq 0.$$

Noting that $\sum_{n=0}^{\infty} \frac{1}{2^n} = 2$, we deduce that

$$\sum_{n=0}^{\infty} \frac{1}{2^n+\sqrt{n}} \leq 2$$

$\Rightarrow$ the series converges

The Limit Comparison Test

Theorem (Limit Comparison Test): Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N an integer).

1. If $\lim_{n\to\infty} \frac{a_n}{b_n} = c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.

2. If $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.

"eventually $a_n \ll b_n \Rightarrow \sum a_n < \sum b_n$, so if the larger series converges, then the smaller series converges"

3. If $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

"eventually $a_n \gg b_n \Rightarrow \sum a_n > \sum b_n \Rightarrow$ if the smaller series $\to\infty$, then the larger series $\to\infty$"

Proof ①: There exists some M such that for $n > M$,

$$\left|\frac{a_n}{b_n} - c\right| < \frac{c}{2} \Rightarrow \frac{c}{2} < \frac{a_n}{b_n} < \frac{3c}{2} \Rightarrow \frac{c}{2} b_n \overset{(*)}{<} a_n \overset{(+)}{<} \frac{3c}{2} b_n$$

If $\sum b_n$ converges $\Rightarrow \sum \frac{3c}{2} b_n$ converges $\overset{\text{DCT + (+)}}{\Rightarrow} \sum a_n$ converges.

If $\sum b_n$ diverges $\Rightarrow \sum \frac{c}{2} b_n$ diverges $\overset{\text{DCT + (*)}}{\Rightarrow} \sum a_n$ diverges.

Examples:

(a). $\sum_{n=1}^{\infty} \frac{2n+1}{(n+1)^2} =$

Check: $a_n \overset{?}{>} \frac{2}{n}$

$\frac{2n+1}{(n+1)^2} > \frac{2}{n}$

$2n^2+n \not> 2n^2+4n+2$

$a_n \overset{?}{>} \frac{1}{n}$

$\frac{2n+1}{(n+1)^2} \overset{?}{>} \frac{1}{n}$

$2n^2+n \overset{?}{>} n^2+2n+1$

$n^2 \overset{?}{>} n+1$

✓ when $n \geq 2$.

$a_n = \frac{2n+1}{(n+1)^2}$ & $b_n = \frac{2}{n}$

then $\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{2n+1}{(n+1)^2} \cdot \frac{n}{2} = \lim_{n\to\infty} \frac{1+\frac{1}{2n}}{(1+\frac{1}{n})^2} = 1$

Since $\sum b_n = \sum \frac{2}{n}$ diverges, so LCT $\Rightarrow \sum a_n$ diverge.

(b). $\sum_{n=1}^{\infty} \frac{1}{2^n-1} =$

Compare with $\frac{1}{2^n}$: $\lim \frac{\frac{1}{2^n-1}}{\frac{1}{2^n}} = 1$

$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{2^n-1}$ converge because $\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$ converges.

(c). $\sum_{n=1}^{\infty} \boxed{\frac{1+n\ln n}{n^2+5}} =$ a_n

Check $\lim_{n\to\infty} \frac{a_n}{b_n} = 1$.

$b_n = \frac{\ln n}{n}$

LCT $\Rightarrow$ $\sum a_n$ & $\sum b_n$ behave similarly in convergence.

To study the convergence of $\sum b_n$, we observe that

$b_n = \frac{\ln n}{n} \geq \frac{1}{n}$ for $n \geq 3$.

Since $\sum_{n=3}^{\infty} \frac{1}{n}$ diverges, so DCT $\Rightarrow \sum b_n$ diverges.

$\Rightarrow \sum a_n$ diverges.

(d). $\sum_{n=1}^{\infty} \frac{\ln n}{n^{3/2}} =$

$a_n = \frac{\ln n}{n^{3/2}}$ Observe that $a_n = \frac{1}{n^{5/4}} \cdot \frac{\ln n}{n^{1/4}}$

Compare with $b_n = \frac{1}{n^{5/4}}$

then $\lim \frac{a_n}{b_n} = \lim \frac{\ln n}{n^{1/4}} = \lim_{x\to\infty} \frac{\ln x}{x^{1/4}} \overset{\text{l'Hôpital}}{=} \lim_{x\to\infty} \frac{1/x}{\frac{1}{4}x^{-3/4}} = 0$

$\sum b_n$ is $\frac{5}{4}$-series, so converges $\overset{\text{LCT}}{\Rightarrow}$ $\sum a_n$ also converges.

Extra Examples:

1. The series $\sum_{n=1}^{\infty} \frac{1}{n^2+n}$ is convergent.
2. The series $\sum_{n=1}^{\infty} \frac{1}{n^2-n+1}$ is convergent.
3. The series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1}}$ is divergent.
4. The series $\sum_{n=2}^{\infty} \frac{(\ln n)^3}{n^{1+\epsilon}}$ is convergent for any $\epsilon > 0$.

(4) $a_n = \frac{(\ln n)^3}{n^{1+\varepsilon}}$, let $b_n = \frac{1}{n^{1+\frac{\varepsilon}{2}}}$

$\left(= \frac{1}{n^{1+\frac{\varepsilon}{2}}} \cdot \frac{(\ln n)^3}{n^{\varepsilon/2}}\right)$

Now since $\sum b_n$ is an $(1+\frac{\varepsilon}{2})$-series, which converges, then by the LCT, $\sum a_n$ converges too.

$$\lim \frac{a_n}{b_n} = \lim_{x\to\infty} \frac{(\ln x)^3}{x^{\varepsilon/2}}$$

$$\overset{\text{l'Hopital}}{=} \lim_{x\to\infty} \frac{3(\ln x)^2 \frac{1}{x}}{\frac{\varepsilon}{2} x^{\frac{\varepsilon}{2}-1}} = \lim \frac{3(\ln x)^2}{\frac{\varepsilon}{2} x^{\varepsilon/2}} = \cdots = 0.$$